Matis Di Piazza

Computer Science Engineering student (4th year) | Polytech Grenoble, Université Grenoble Alpes
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Portfolio: matisdpz.github.io | GitHub: github.com/matisdpz
Seeking an internship | Available: April-August 2027 (full-time) | Expected graduation: September 2028

Education

INP Grenoble - Polytech Grenoble - Université Grenoble Alpes **Grenoble, France**
Master's degree in Computer Science Engineering (Diplôme d'ingénieur) 2025-2028

- **Coursework:** software architecture, object-oriented programming, databases and data modeling, computer networks, IT infrastructure, embedded systems, data analysis

Polytech Lyon - Université Claude Bernard Lyon 1 **Lyon, France**
Classes préparatoires *(intensive 2-year course in advanced Maths, Physics & Computer Science)* 2023-2025

Projects

AI SRE Assistant | Python, FastAPI, PostgreSQL, React, TypeScript, Docker | github.com/matisdpz/ai-sre-assistant **2026**

- Built an incident management platform for a simulated distributed system: metrics ingestion API, rule-based incident detection and a React dashboard (MTTA, MTTR)
- Designed concurrency-safe incident deduplication (row locks, partial unique index) and idempotent ingestion with retries
- Set up CI with GitHub Actions: 164 backend tests including integration tests on PostgreSQL, migration drift check, Docker builds
- Integrated an optional LLM (Google Gemini) that suggests hypotheses

Music-Theory app | Java, Spring Boot, PostgreSQL, React, TypeScript, Docker | github.com/matisdpz/harmony-lab **2026**

- Built a full-stack music-theory app: progression editor, Roman-numeral analysis, interactive piano with audio
- Designed a framework-free Java music-theory engine with voice-leading optimization by dynamic programming
- REST API (Spring Boot, JPA, Flyway, optimistic locking); CI with JUnit, Testcontainers, Vitest, Docker builds

AI/Robotics Simulation | Python

- Designed a 2D state-space environment for a drone; implemented a recursive state exploration algorithm to validate the environment before integrating a learning agent

Other Projects | C, Java, C++

- Client-server file transfer application (TCP/IP, Unix sockets); embedded C software for an STM32 microcontroller; object-oriented video games in Java and C++

Technical Skills

- **Programming:** Java, C/C++, Python, SQL, TypeScript, OCaml, R, PHP, VBA
- **Backend & data:** Spring Boot, FastAPI, React, REST APIs, PostgreSQL, JPA/Hibernate, SQLAlchemy, Flyway, data analysis, statistics, Tableau, Power BI
- **DevOps & tools:** Git, GitHub Actions (CI), Docker, Docker Compose, Linux, pytest, JUnit, Testcontainers, Vitest
- **Systems & networking:** TCP/IP, Unix sockets, computer networks, embedded C (STM32)
- **AI/ML:** reinforcement learning (MDPs, Q-learning, SARSA), state-space modeling, LLM API integration

Professional Experience

AXA Partners, Business Line Truck | Intern, Actuarial Team | Lyon **February 2023 - March 2023**

- Extracted and processed data from a data lake to support actuarial analysis
- Improved and automated a profitability analysis tool (Excel/VBA)
- Automated reporting workflows and built KPI dashboards for decision support

Hospices Civils de Lyon | Cardiological Hospital, Pharmacy | Lyon **August 2026**

- Performed daily conformity checks (air pressure, decontamination) in a controlled-atmosphere zone, with systematic documentation per Good Manufacturing Practices (GMP)

Community Social Services Center of Saint-Priest | Lyon **January 2023**

- Collected, cleaned and analyzed survey data; produced statistical reports for decision support

Languages & Interests

- **Languages:** French (native), English (B2), Spanish (B2)
- **Interests:** piano (12 years in music school), music composition, strategy games